M-REGULARITY OF A COHERENT SHEAF ON PROJECTIVE SPACE IS PRESERVED WHEN M INCREASES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The induction step producing a section with a hyperplane restriction. Stacks Project, Varieties, Lemma lemma-exact-sequence-induction (tag 0ABH). Supplies the short exact sequence used in the induction step and the sheaf G on the hyperplane.
- Base-field invariance of m-regularity and the restriction lemma. Stacks Project, Varieties, Lemmas lemma-m-regular-extend-base-field (tag 0A2A) and lemma-m-regular (tag 0A2B). The first allows enlarging the field and the second gives that the restriction G is m-regular.

Source reference. Original lemma 08A4: sources/stacks-chow/tags.txt, source lines 10519–10519. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 08A0: sources/stacks-chow/tags.txt, source lines 10515–10515. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 08A5: sources/stacks-chow/tags.txt, source lines 10520–10520. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original remark 0EGK: sources/stacks-chow/tags.txt, source lines 18100–18100. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let k be a field. Let $n \geq 0$. Let $\mathcal{F}$ be a coherent sheaf on $\mathbf{P}^n_k$. If $\mathcal{F}$ is m-regular, then $\mathcal{F}$ is $(m+1)$-regular.*

Proof. We prove this by induction on n. If $n = 0$ every sheaf is m-regular for all m and there is nothing to prove. By Lemma 08A4 we may replace k by an infinite overfield and assume k is infinite. Thus we may apply Lemma 08A0. By Lemma 08A5 we know that $\mathcal{G}$ is m-regular. By induction on n we see that $\mathcal{G}$ is $(m+1)$-regular. Considering the long exact cohomology sequence associated to the sequence

$$0 \to \mathcal{F}(m-i) \to \mathcal{F}(m+1-i) \to i_*\mathcal{G}(m+1-i) \to 0$$

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as in Remark 0EGK the reader easily deduces for $i \geq 1$ the vanishing of $H^i(\mathbf{P}^n_k, \mathcal{F}(m+1-i))$ from the (known) vanishing of $H^i(\mathbf{P}^n_k, \mathcal{F}(m-i))$ and $H^i(\mathbf{P}^n_k, \mathcal{G}(m+1-i))$. $\square$